BHUVAN M ACHARYA

+91 96066 54776 | bhuvanm200623@gmail.com | github.com/j4b3-21 | linkedin.com/in/bhuvan-m-acharya-7b9256335

Education

Indian Institute of Information Technology Tiruchirappalli — Tiruchirappalli, India
B.Tech. in Computer Science and Engineering | CGPA: 8.5/10 — *2024 – 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Machine Learning

Experience

Nebula Technologies Labs Pvt. Ltd. — Remote
Associate Software Engineer Intern / Founding Engineer — *May 2026 – Aug. 2026*

- Architected and developed a production CRM platform from the ground up using **Java, Spring Boot, React, PostgreSQL, and REST APIs**, owning backend architecture and end-to-end feature development.
- Designed production infrastructure using **Docker, Nginx, VPS hosting, and Cloudflare**, focusing on service isolation, reliability, deployment, and maintainability.
- Built automated **CI/CD pipelines with Jenkins and GitHub Actions**, reducing deployment time from approximately **10 minutes to under 1 minute**.
- Implemented and tested backend services using **JUnit, Postman, and Bruno**; debugged production issues involving APIs, database failures, backend exceptions, logs, and deployment failures.
- Worked with **PostgreSQL** including indexing, query optimization, transactions, isolation levels, migrations, JSONB, connection pooling, and database design.
- Developed concurrent and asynchronous backend components using **Java concurrency, thread pools, executors, concurrent collections, message queues, and RabbitMQ/Kafka**.
- Collaborated in a **6-member engineering team** across technical design, architecture discussions, code reviews, Jira-based issue tracking, debugging, release workflows, and production deployments.
- Maintained Git/GitHub repositories and engineering workflows while working across an evolving codebase and incorporating feedback from engineers and stakeholders.

Indian Institute of Information Technology Tiruchirappalli — Tiruchirappalli, India
Undergraduate Research Collaborator — *2026 – Present*

- Developing an open-source graph-native platform for reproducible computational research, focusing on experiment traceability and research workflow infrastructure.
- Designing systems for **dataset versioning, experiment tracking, workflow orchestration, artifact management, and experiment lineage**.
- Contributing to implementation and evaluation methodologies for reproducible computational neuroscience and fMRI-based research.

Projects

Priority Tag File System — xv6 Kernel — *C / xv6 / QEMU*

- Modified the **MIT xv6 kernel** to implement a user-driven file system that dynamically places files according to user-defined priority tags and access frequency.
- Extended the kernel with new **system calls, tag-management modules, inode metadata, priority calculation, and priority-based block allocation**.
- Implemented a weighted priority model combining base priority, tag weights, and access frequency, followed by dynamic block reordering based on computed priority.
- Built kernel/user-space utilities including **tag, addtag, removetag, listtags, and dumpblocks** to manage priorities and inspect block allocation.
- Tested the modified kernel using **QEMU**, validating dynamic block reordering, priority assignment, tag metadata, and compatibility with existing xv6 file operations.

CampusHire — Placement CRM & ERP Platform — *Java / Spring Boot / React / PostgreSQL*

- Developed a full-stack enterprise platform with **REST APIs, configurable workflows, role-based access control, secure authentication, and modular backend services**.
- Architected a metadata-driven backend supporting dynamic forms, records, layouts, relationships, and configurable business processes using PostgreSQL.
- Designed backend components using **authentication/authorization, database architecture, asynchronous processing, caching, and workflow orchestration**.
- Containerized services and automated deployment using **Docker, Jenkins, and Nginx**.

NeuroWeave — Graph-Native Research Platform — *Python / React / PostgreSQL*

- Building an open-source platform for reproducible computational research using graph-based workflow management and experiment lineage.
- Developing experiment tracking, artifact versioning, dataset management, and workflow orchestration capabilities for scientific computing environments.

Technical Skills

Languages: Java, C++, Python, Go, JavaScript, TypeScript, SQL, C

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, Concurrency, Distributed Systems

Backend: Spring Boot, Node.js, Express.js, REST APIs, gRPC, Microservices, Event-Driven Architecture, Asynchronous Processing

Databases: PostgreSQL, MySQL, Redis, Vector Databases, Indexing, Query Optimization, Transactions, ACID, Isolation Levels, JSONB

Concurrency & Messaging: Java Threads, ExecutorService, Thread Pools, Locks, Concurrent Collections, RabbitMQ, Kafka, Message Queues

Cloud & Infrastructure: Docker, Kubernetes, Jenkins, GitHub Actions, Nginx, AWS, Linux, VPS, Cloudflare

Testing & Engineering: JUnit, Postman, Bruno, Git, GitHub, CI/CD, Debugging, Code Reviews, Jira, System Design

Achievements & Open Source

Competitive Programming: Solved **300+ problems on LeetCode**; achieved a maximum **Codeforces rating of 1180**; participated in college programming contests.

Open Source: Active open-source contributor with public GitHub projects and merged pull requests; applied twice to Google Summer of Code and actively contributes to open-source software.

Leadership & Campus Involvement

Training & Placement Representative IIIT Tiruchirappalli
Student Placement Representative *2025 – Present*

- Coordinated recruiter, student, and placement operations while contributing to the development of the institution's placement CRM/ERP platform.

Head — Web Development Group IIIT Tiruchirappalli
Web Development Lead *2025 – Present*

- Mentored student developers and contributed to maintaining institutional web applications and college websites.

Certifications

Deep Learning Specialization — DeepLearning.AI 2025–2026

Machine Learning Specialization — Stanford University & DeepLearning.AI 2025

Natural Language Processing — DeepLearning.AI 2026